T2D Glucose Digital Twin — System Architecture

Page 1: Scope, Multimodal Ingestion Contract, & Hybrid Physiological-Deep Learning Engine

System Boundary & Hypothesis: Investigational engineering prototype combining observable patient context, wearable history, and mechanistic physiological modeling to forecast glucose trajectories and adverse spike risks over a 120-minute horizon. Evaluated on synthetic physiological cohorts (293 subjects); not clinically validated for autonomous medical decisions.

TIER 1 Multimodal Observable Ingestion (Contract S3) & Latent Firewall

Privileged generator internal latents (e.g., secret true insulin kinetics) are strictly quarantined by the latent firewall. The system admits only clinically observable streams:

Static EHR Context
Age, BMI, HbA1c, eGFR, T2D duration, 7 medication class flags (metformin, SGLT2i, GLP-1 RA, DPP-4i, sulfonylurea, basal and prandial insulin).

Dynamic CGM Telemetry
5-minute sampled continuous glucose, ROC trend velocity (Δmg/dL/min), sensor flag, warm-up state.

Wearable Biometrics
Continuous Heart Rate (HR), HRV, step counts, sleep stages, and resting HR warm-up baselines.

Dietary & Behavioral Logs
Logged meal carbohydrates (grams), meal timing, absorption slots, and population habit priors.

TIER 2 Hybrid Physiological Prior + Deep Residual Learning

Mechanistic UKF S3 Engine:

Two-compartment Bergman minimal model coupled to a two-stage nonlinear gut absorption ODE. Parameterized by patient-specific EHR priors:

$$\begin{aligned} dG/dt &= -(p_1 + X)G + p_1 G_b + Ra(t) / V_G \\ dX/dt &= -p_2 X + p_3 (I - I_b) \end{aligned}$$

Unscented Kalman Filter (UKF) dynamically tracks insulin sensitivity (S_I) and gastric emptying (t_{max}), delivering continuous 120-minute mechanistic forward rollouts.

PR-TFT Deep Residual Architecture:

Physics-Residual Temporal Fusion Transformer (PR-TFT) learns behavioral and physiological residuals over the mechanistic baseline:

$$\hat{y}_t^{(q)} = \hat{y}_t^{\mathrm{UKF}} + \hat{r}_t^{(q)}(\text{observable history, context})$$

Variable Selection Networks (VSN) filter multimodal inputs; temporal LSTM encodes past dynamics; causal self-attention models long-range multi-horizon dependencies.

Residual Quantile Heads (Multi-Horizon)

Outputs 7 quantiles (**p05, p10, p25, p50, p75, p90, p95**) across 24 discrete 5-minute steps (120 min total). Conformal adjustment brings the 80% interval to 80.00% empirical coverage on the validation partition (a resubstitution diagnostic, not a guarantee).

Discrete-Time Spike Hazard Head

Predicts cumulative onset probability *P*(spike within 120 min). Calibrated post-training via isotonic regression (ECE 0.000094, a resubstitution diagnostic on the validation partition); it drives research alert tiers, not clinical alerts.

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Page 2: Reliable Telemetry Serving, Chunk-Invariant SSE Proxy, & Clinical Workstation

TIER 3 Reliable Inference Serving & Telemetry Proxy Architecture

FastAPI Inference Service (Port 8001)
backend/serve_pr_tft.py serves Blueprint Section 9 JSON contract. Evaluates data-quality gates, runs UKF rollouts + PR-TFT inference, checks quantile monotonicity, and executes stateful AlertMachine with 120-min refractory suppression.

Next.js SSE Proxy (Port 3000)
src/app/api/patients/[id]/stream/route.ts proxies events using createSafeSSETransformStream. Chunk-invariant parsing searches for earliest delimiter (\r\n\r\n, \n\n, \r\r), immune to arbitrary network framing and TCP fragmentation.

Client Hook (useRealtimeTelemetry)
State machine manages connection lifecycles: connecting → connected → reconnecting. Bounded exponential backoff (max 3 retries), envelope schema validation via Zod, and 15-second client staleness detection.

TIER 4 Clinical Workstation Interface

Cohort Directory
/cohort page. Demographic search, status filters (Healthy, Prediabetes, T2D), and quick-pivot patient cards.

Patient Workspace
/patients/[id]. Visualizes replayed synthetic CGM, 24-step forecast ribbon, sensor staleness, and twin parameters.

Risk Panel & Alert Strip
Cumulative adverse event hazard curve with research threshold badges (Watch at $p \geq 0.35$, Warning at $p \geq 0.60$).

Model Contributions
Additive decomposition explaining model arithmetic: mechanistic prior, meal absorption, learned residual.

SAFETY BOUNDARY Scenario Exploration (Interactive What-If)

Supported (UI → /api/forecast → /v1/forecasts):
Routes /api/forecast and /v1/forecasts accept hypothetical meal additions/skips and physical activity inputs, re-solving the linearized twin and re-scoring with PR-TFT from the same anchor. The patient workspace shows baseline vs. scenario ribbons, highest median, largest same-time difference and Δprobability.

Clinical Boundary (Enforced):
Every result is labelled “Simulated model response — not a validated causal treatment effect.” Pharmacological scenarios return SCENARIO_UNSUPPORTED (422). Counterfactual treatment effects have not been validated in humans and must not guide therapy.

Fail-Closed Safety Architecture: If the neural network fails (OOM, timeout), the service falls back to the mechanistic UKF twin-only trajectory (fallback_used: "twin_only") with monotonic quantiles and suppresses hazard alerts. If CGM coverage is insufficient (<6h), forecast and risk are strictly nulled.

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Page 3: Patient-Specific Hybrid Synergies, Calibrated Uncertainty, & Scientific Limitations

CORE DIFFERENTIATION

Why a Physics-Residual Hybrid Architecture?

- **Mass Conservation & Physiological Grounding:** Pure black-box neural networks frequently hallucinate impossible glucose spikes or cliff-drops. The UKF-tracked minimal model keeps the twin on glucose–insulin dynamics; the learned adjustment can still add isolated steps, which the product flags.
- **Patient-Specific Personalization:** UKF continuously tracks individual metabolic traits (EHR priors: insulin sensitivity S_I and absorption time t_{max}) from longitudinal history.
- **Deep Residual Learning:** The PR-TFT transformer does not need to re-learn basic human physiology; it focuses entirely on modeling complex residual behaviors (delayed dawn phenomenon, stress, meal composition).
- **Interpretability for Clinicians:** Decomposes predictions into observable physiological drivers and learned corrections without black-box opacity.

EVIDENCE

Validation Benchmark & Calibrated Uncertainty

- **Evaluated Validation Partition:** 44 validation patients (54,281 eligible anchors; 47,629 is the at-risk subset used for hazard calibration) from pre-registered split_293.json.
- **Trajectory Error (RMSE mg/dL):**
  - **30-min:** 12.36 (MAE 8.43, MARD 7.22%)
  - **60-min:** 19.53 (MAE 12.43, MARD 10.16%)
  - **120-min:** 25.03 (MAE 16.76, MARD 13.66%)
- **Conformal Uncertainty:** 80% empirical interval coverage is **80.00%** (mean width 37.3 mg/dL).
- **Hazard Calibration:** Isotonic-calibrated spike hazard achieves AUROC **0.9226**, AUPRC **0.6314**, and Expected Calibration Error (ECE) **0.000094**.

SCIENTIFIC INTEGRITY

Negative Validation Gates & Explicit Limitations

1. Gate 3 Realism Audit: FAILED

Historical AUROC **0.9320** / Duration-corrected AUROC **0.9290** vs. threshold **≤ 0.65**. Discriminator easily separates synthetic ODE series from mock-real CGM (top tell: jerk_sd). Reported factorial drift controls (scripts not in this repository) indicate generator AR(1) drift explains part of curvature differences, but not the full multivariate failure. No fabricated smoothing applied.

2. Baseline Audit: UNVERIFIED

Demo baseline files (Persistence RMSE120 50.66, UKF 40.20) evaluated an 18-patient test subset under an earlier demo run. Because populations are not aligned, direct superiority claims on the 44-subject validation split are marked cohort-unverified.

3. EW-01 Protocol: UNEXECUTED

Formal comparative early warning evaluation remains governance-blocked pending co-author sign-offs. The 44-patient validation split was used for model tuning and calibration; it does not represent independent human generalization.

Investigational Research Prototype: Engineering checks: 45 Jest suites, 424 tests; 140 backend tests. Negative findings, including the failed realism gate, are published with the results.